AI MENTAL HEALTH UK

Task - Data documentation & metadata

Duration – 6-9 May

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Domain - Data Science

1. Introduction

This document provides metadata, dataset documentation, provenance information, and NHS data compliance notes for the Adult Psychiatric Morbidity Survey (APMS) 2014 dataset used throughout the project.

The purpose of DS-4 is to ensure:

- dataset transparency,
- reproducibility,
- ethical usage,
- and proper documentation of variables and sources.

The project used publicly available APMS 2014 NHS Digital statistical tables.

2. Dataset Overview

Attribute	Description
Dataset Name	Adult Psychiatric Morbidity Survey (APMS) 2014
Publisher	NHS Digital
Research Organisation	NatCen Social Research
Geographic Coverage	England
Population	Adults aged 16+
Survey Year	2014
Total Respondents	7,546
Data Type	Aggregated statistical survey tables
File Format	Excel (.xls)

3. Files Used in Project

File	Purpose
Chapter 2 CMD Tables	Anxiety, depression, CMD prevalence
Chapter 4 PTSD Tables	PTSD prevalence and trauma analysis
Chapter 12 Tables	Suicidal thoughts and self-harm
Chapter 14 Tables	Confidence intervals and statistical methods

4. Dataset Provenance

The APMS 2014 survey was conducted by NHS Digital in collaboration with NatCen Social Research.

The survey was designed to estimate:

- prevalence of mental disorders,
- demographic mental health patterns,
- regional variation,
- and treatment-related outcomes among adults in England.

The data used in this project was extracted directly from NHS statistical tables and not from third-party sources.

No manually created prevalence values were used during analysis.

5. Dataset Dictionary

Core Variables

Variable	Description
age10yr	Age grouped into 10-year bands
ressex	Respondent sex
anycmd	Any common mental disorder
gad	Generalised Anxiety Disorder
dep	Depressive episode
phob	Phobia
ocd	Obsessive compulsive disorder

Variable	Description
panic	Panic disorder
cmd_nos	CMD not otherwise specified
gor	Government Office Region
PTSD_COMB	PTSD screen-positive indicator
SuiThComb	Suicidal thoughts
SuiAtComb	Suicide attempts
shevcomb	Self-harm indicator

6. CIS-R Severity Bands

The Clinical Interview Schedule – Revised (CIS-R) was used to measure psychological distress severity.

Score Range	Interpretation
0–5	Minimal symptoms
6–11	Mild symptoms
12–17	Moderate CMD
18+	Severe CMD

7. Data Processing Notes

The following preprocessing steps were performed during analysis:

- Extracted prevalence values using pandas.
- Converted percentage values into numeric format.
- Cleaned missing values using errors='coerce'.

- Generated prevalence distributions for analysis.
- Reconstructed pseudo severity scores for DS-3 distribution analysis.

Only aggregated statistical data was processed.

No respondent-level identifiable information was accessed.

8. NHS Data Usage Compliance Checklist

Compliance Requirement	Status
Public NHS dataset used	✅
No personal identifiable data used	✅
Educational/research use only	✅
NHS attribution maintained	✅
Aggregated statistical data only	✅
No individual identification attempts	✅
Ethical analytical use maintained	✅

9. Ethical Considerations

Mental health datasets contain sensitive health-related information. Although APMS 2014 tables are publicly available and aggregated, responsible interpretation remains important.

The project:

- avoided individual-level identification,
- maintained NHS attribution,
- and used the dataset only for analytical and educational purposes.

10. Reproducibility

The project was implemented using:

- Python
- Pandas
- NumPy
- Matplotlib
- SciPy

All analyses are reproducible using:

- APMS Excel tables,
- documented extraction methods,
- and notebook workflows.

11. Conclusion

The DS-4 documentation provides structured metadata and provenance information for the APMS 2014 dataset used throughout the project.

The report supports:

- reproducibility,
- ethical data usage,
- transparency,
- and proper interpretation of mental health prevalence analyses.